

Homework 7

Sections 2 and 3: MWF 9:30am–11:20am (Rosenwald 011)

Spring 2026

Please submit your homework pdf on Gradescope by 6pm on Wednesday, May 13.

Reminder: Your method of reasoning is important, even if your final answer is incorrect. Where a numerical answer is requested, please show how you arrived at your final answer. If you refer to a plot, make sure the plot is made clearly visible alongside your answer in your homework writeup.

Problem 1

The melting point of each of $n = 16$ samples of a brand of hydrogenated vegetable oil was determined, resulting in $\bar{X}_n = 94.32$. Assume that the distribution of melting point is normal with $\sigma = 1.20$. Let μ_1 denote the (unobserved) true melting point for this brand of oil.

- (a) Test $H_0 : \mu_1 = 95$ versus $H_1 : \mu_1 \neq 95$ using a two-sided level 0.01 test.
- (b) Based on your conclusion in part (a), would a 99% confidence interval for the (true) melting point for this brand of hydrogenated vegetable oil contain the value 95?

Suppose now that a 95% confidence interval for a parameter μ_2 is $(3.1, 5.1)$ and the 99% confidence interval for μ_2 is $(2.8, 5.4)$. For parts (c) and (d), consider testing between the hypotheses $H_0 : \mu_2 = 3$ versus $H_1 : \mu_2 \neq 3$. Please answer the following questions and briefly explain your reasoning.

- (c) Would H_0 be rejected at significance level $\alpha = 0.05$?
- (d) Would H_0 be rejected at significance level $\alpha = 0.01$?

For parts (e) and (f), suppose we now wish to test $H_0 : \mu_3 = 10$ versus $H_A : \mu_3 \neq 10$, and we observe a two-sided p -value of $p = 0.03$. Please answer the following questions and briefly explain your reasoning.

- (e) Does the 95% confidence interval for μ_3 include 10?
- (f) Does the 99% confidence interval for μ_3 include 10?

Problem 2

The Marist Poll published a report stating that 66% of adults nationally think licensed drivers should be required to retake their road test once they reach 65 years of age. It was also reported that interviews were conducted on 1,018 American adults, and that the margin of error was 3% using a 95% confidence level. For parts (a) and (b), please indicate if the statements are true or false and explain your reasoning.

- (a) We are 95% confident that 63% to 69% of the 1018 American adults surveyed thought licensed drivers should be required to retake their road test once they turn 65.
- (b) We are 95% confident that 63% to 69% of American adults thought licensed drivers should be required to retake their road test once they turn 65.

For parts (c) – (e), we will conduct a hypothesis test to determine whether 60%, **or more** than 60% of the population thought that licensed drivers should be required to retake their road test once they turn 65. Let p denote the true proportion of adults in the population who hold this belief.

- (c) Write down the null and alternative hypotheses (these should be in terms of p).
- (d) Formulate a test statistic for deciding between H_0 and H_1 , and state its null distribution.
- (e) Report the p -value, and make a conclusion at the 0.01 significance level.

Problem 3

The proportion of drivers who use seat belts can differ widely between sub-populations grouped by age, gender, or local law. At the time of a research study, in New York City, police could stop a driver who is not belted, whereas in Boston, police could cite a driver for not wearing a seat belt only if the driver had been stopped for some other violation. Here are data from observing random samples of female Hispanic drivers in these two cities.

City \ Seatbelt?	Yes	No	Total
New York City	183	37	220
Boston	68	49	117

- (a) Verify that the 95% CI for the difference in the proportions of female Hispanic drivers in the two cities who wear seat belts $p_{\text{NY}} - p_{\text{BT}}$ is (0.148, 0.353). Show your computation.

- (b) Determine if the following statements are true or false, and explain your reasoning if you claim a statement is false.
- (i) There was strong evidence that $p_{\text{NY}} > p_{\text{BT}}$.
 - (ii) 95% of random samples of 220 female Hispanic drivers in NYC and 117 female Hispanic drivers in Boston will yield differences in sample proportions between 14.9% and 35.3%.
 - (iii) A 90% confidence interval for $p_{\text{NY}} - p_{\text{BT}}$ would be wider than the interval in part (a).
- (c) Test the hypothesis that the population proportions are the same, i.e. $H_0 : p_{\text{BT}} = p_{\text{NY}}$ versus $H_A : p_{\text{BT}} \neq p_{\text{NY}}$ at the $\alpha = 0.01$ significance level. Specifically, please formulate a test statistic and state its null distribution. Finally, compute the statistic for this data, along with a p -value and state your conclusion within the context of this study.

Problem 4

Suppose that a random sample is selected from an approximately normal population with an unknown standard deviation. For each of the given sets of hypotheses, sample size, and t -statistic, please find the p -value. If you use R, please provide the line of code used to generate the p -value.

- (a) $H_0 : \mu = 0.5$, $H_1 : \mu > 0.5$, $n = 26$, $T = 2.6$.
- (b) $H_0 : \mu = 0.5$, $H_1 : \mu \neq 0.5$, $n = 26$, $T = 2.6$.
- (c) $H_0 : \mu = 3$, $H_1 : \mu < 3$, $n = 18$, $T = -2.2$.
- (d) $H_0 : \mu = 3$, $H_1 : \mu < 3$, $n = 18$, $T = 2.2$.

Problem 5

The nominal percentage of ammonia in a certain gas produced at a chemical plant is 12%, but the actual value fluctuates daily. The ammonia content was measured by two technicians on 9 successive days, and the following deviations from 12% were recorded¹. The recorded values are $100 \cdot (\text{concentration} - 12)$, so for example, the value 48 means that the ammonia concentration was measured to be 12.48%:

```
tech.A <- c(4, 37, 35, 43, 34, 36, 48, 33, 23)
tech.B <- c(18, 37, 38, 38, 47, 48, 57, 28, 32)
```

¹This data is taken from O.L. Davies: *Statistical Methods in Research and Production*.

- (a) Calculate the sample mean and sample variance of the values for each technician.
- (b) The daily differences are defined $D_t = Y_{B,t} - Y_{A,t}$. Compute the sample mean of the differences, and the sample variances of the differences.
- (c) Test the hypothesis that there is no systematic difference between the values obtained by the two technicians. State the null hypothesis, the alternative hypothesis, and compute the two-sided p -value. What do you conclude from this?
- (d) Compute a 95% confidence interval for the difference $\mu_B - \mu_A$ of the means for the two technicians.

Problem 6

This question is related to the analysis of data collected in a series of 15 experiments by the Scottish chemist William Ramsay and the English physicist John William Strutt in 1894. Samples of Nitrogen gas (N_2) were carefully prepared, dried, and weighed. Seven samples were derived from atmospheric sources, and eight from chemical sources. The weights obtained for equal volumes of gas were as follows.

```
at.n2 <- c(2.31017, 2.30986, 2.31010, 2.31001, 2.31024, 2.31010, 2.31028)
ch.n2 <- c(2.30143, 2.29890, 2.29816, 2.30182, 2.29869, 2.29940, 2.29849, 2.29869)
```

Please complete parts (a) through (e), without using the R function `t.test()`.

- (a) Calculate the sample mean and sample variance for atmospheric N_2 . Additionally, report a 99% confidence interval for the mean weight of the given volume of atmospheric N_2 .
- (b) Repeat part (a) for the chemically sourced N_2 .
- (c) Compute the pooled sample variance, using all 15 measurements. How many degrees of freedom are attached to the pooled variance?
- (d) Use the pooled sample variance to construct a 99% confidence interval for the mass difference (atmospheric N_2 minus chemical N_2).
- (e) The degrees of freedom for the Welch two-sample t -test or confidence interval are

$$d = \frac{(n_1 - 1)(n_2 - 1)(n_2 S_1^2 + n_1 S_2^2)^2}{n_2^2(n_2 - 1)S_1^4 + n_1^2(n_1 - 1)S_2^4}$$

where n_1, n_2 are the sample sizes, and S_1^2, S_2^2 are the sample variances. Without pooling variances, use the above formula for the Welch degrees of freedom to construct a 99% confidence interval for the mass difference (atmospheric – chemical). Report the degrees of freedom, the relevant t -quantile for a 99% interval, and the interval itself.

(f) Verify your computations in parts (a), (b), (d) and (e) using the R function `t.test()`:

```
t.test(at.n2, alternative = "two.sided", conf.level=0.99)
t.test(ch.n2, alternative = "two.sided", conf.level=0.99)
t.test(at.n2, ch.n2, var.equal=TRUE, conf.level=0.99)
t.test(at.n2, ch.n2, var.equal=FALSE, conf.level=0.99)
```

Briefly summarize your conclusion from this data regarding the comparison between the molecular weights of atmospheric versus chemical N₂.